

Assignment 3

Your Name Here

A microbe called *Cinderspore* is spreading between star systems and dimming stars as it goes. One research vessel, the *Meridian*, was sent out with a small crew to study it and find a way to stop it before it reaches Earth.

The *Meridian*'s crew is busy keeping the ship running and the mission alive while you are analyzing the data sent by Frank. We have a new transmission:

Mission Log 003 — from Frank

Okay Analyst, hi, good, you're here. Last time you told us what was *normal* up here — the averages, the typical spread, all that. Real helpful! Made everyone feel very Seen. But Mission Control keeps asking me a different kind of question now, and it's one number can't answer alone. It's always "compared to what?"

Is Flight Ops actually breathing different air than Support crew, or does it just feel that way when you've been awake for thirty hours? Did the quiet-hours thing I begged Command for actually help anybody sleep, or did I just annoy everyone for nothing? Is one end of this ship really less reliable than the other, or am I just superstitious about the Aft airlock?

Variable	Type	Description
crew_role	Categorical	Pilot, Engineer, Medic, or Analyst
duty_track	Categorical (2 levels)	"Flight Ops" (Pilot, Engineer) or "Support" (Medic, Analyst)
ship_module	Categorical	Where the reading was logged: Engineering, Med Bay, Science Deck, Cargo Bay
zone	Categorical (2 levels)	"Forward Zone" (Engineering, Science Deck) or "Aft Zone" (Med Bay, Cargo Bay)
mission_phase	Categorical	Transit, Orbit, or Planetside
cinderspore_growth_rate	Continuous	Estimated microbe growth rate (spores/hr)

Variable	Type	Description
crew_sleep_hours	Continuous	Hours slept in the last rest cycle, logged in Week 1
crew_sleep_hours_week3	Continuous	The <i>same</i> crew member's hours slept, re-logged in Week 3, after the quiet-hours policy started
hull_temp_f	Continuous	Hull temperature (°F)
o2_reserve_pct	Continuous	Oxygen reserve remaining (%)
o2_alert	Categorical (2 levels)	“Alert” if o2_reserve_pct dropped below 70%, otherwise “Normal”

Question 0: Data Import

0a. Call in any packages that are needed for this assignment.

0b. Import this week's data here.

Question 1: Flight Ops vs. Support – Cinderspore Readings

Frank: “Flight Ops is in and out of Engineering and the Science Deck constantly. Support mostly stays put in Med Bay. I keep wondering if that's why their Cinderspore growth-rate readings *feel* different when I skim the log. Is it real?”

Question: Does the typical Cinderspore growth rate (cinderspore_growth_rate) differ between Flight Ops and Support crew (duty_track)?

1a. Before running any tests, construct the appropriate QQ plot for Cinderspore growth rate.

1b. Based on the graph above, do you believe that we meet the assumption for the appropriate *t*-test? You must explain why or why not.

Your answer here.

1c. Which approach will you use for inference? The two-sample *t*-test or the Wilcoxon rank sum test? Why is that your choice?

Your answer here.

1d. Based on your answer in 1c, find the appropriate point and interval estimates for the difference in Cinderspore growth rate.

1e. Based on your answer in 1c, construct the appropriate hypothesis test to determine if the typical Cinderspore growth rate differs between Flight Ops and Support crew.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

Question 2: Whose Sleep Is Less Predictable?

Frank: "Here's a thing that keeps me up at night. Flight Ops either sleeps great or barely at all, depending on the shift. Support seems more... consistent? Command wants to know if Flight Ops sleep is genuinely *more variable*, not just lower on average."

Question: Is the variability in amount of sleep (`crew_sleep_hours`) greater for Flight Ops than for Support (`duty_track`)?

2a. Find the appropriate point and interval estimates for the variability in amount of sleep.

2b. Construct the appropriate hypothesis test to determine if the variability in amount of sleep is greater for Flight Ops than for Support.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

Question 3: Did Quiet Hours Actually Help?

Frank: “Three weeks ago I convinced Command to try ‘quiet hours’ — no comms, no drills, no me, for eight hours a cycle. I had everyone log their sleep again in Week 3 so we could check. Please tell me it worked. Please.”

Question: Did crew sleep hours increase between Week 1 (`crew_sleep_hours`) and Week 3 (`crew_sleep_hours_week3`) for the same crew?

3a. Before running any tests, construct the appropriate QQ plot for the difference between weeks 1 and 3.

3b. Based on the graph above, do you believe that we meet the assumption for the appropriate t -test? You must explain why or why not.

Your answer here.

3c. Which approach will you use for inference? The paired t -test or the Wilcoxon signed rank test? Why is that your choice?

Your answer here.

3d. Based on your answer in 3c, find the appropriate point and interval estimates for the difference in crew sleep hours.

3e. Based on your answer in 3c, construct the appropriate hypothesis test to determine if crew sleep hours increased between weeks 1 and 3.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...

Question 4: Is One End of the Ship Less Reliable?

Frank: “Okay, last one, and this one’s personal. I think the Aft Zone’s O2 scrubbers are older and worse and I have said this in three separate meetings. Command says I’m being dramatic. Prove me right. Or, fine, prove me wrong. I can take it. Probably.”

Question: Does the proportion of O2 alerts (o2_alert) differ between the Forward Zone and the Aft Zone (zone)?

4a. Find the appropriate point and interval estimates for the proportion of O2 alerts.

4b. Construct the appropriate hypothesis test to determine if the proportion of O2 alerts differ between the two zones.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p -value

- *statistic* =
- p =

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0 .
- There is or is not sufficient evidence to conclude...